Title: Autonomous Traffic Signal Control via Reinforcement Learning

Abstract

The congestion in urban traffic at signalized intersections contributes to delay, fuel consumption and emission, thus, prompting the use of adaptive control over fixed-time and conventional actuated plans. This thesis uncovers the study of autonomous traffic signal control using reinforcement learning (RL) and covers the emerging evidence on multi-agent reinforcement learning (MARL) to coordinate junction management. Basing evaluation on a synthesis of secondary-data, three controllers were compared tabular Q-Learning, Deep Q-Networks (DQN), and a hybrid Q-Learning with a Small Language Model (SLM) advisor, which did so at regular intervals to generate congestion labels, recommendations, and explanations. Findings revealed that even though there are obvious variations in learning dynamics, action preferences, and reward paths, mean evaluation waiting time was effectively the same across models (stabilizing to 29.46 s). DQN along with step-capping Q-Learning showed consistent single-action failure, whereas step-capping Q-Learning encouraged short-horizon conservative behavior. The higher interpretability of the SLM-enhanced agent, which produced decision traces that can be read by humans, came at a very marginal tradeoff in training variance and no delay improvement. Overall, the results emphasize the need to match the reward design, the state representation and the training-evaluation schedule and encourage further research regarding discriminative rewards, network-scale MARL coordination, and deterministic sim-to-real validation in the presence of realistic shifts in demands in general.

Keywords: Autonomous Traffic Signal Control, Reinforcement Learning, Multi-Agent Reinforcement Learning, SUMO, Q-Learning, Deep Q-Network, Reward Shaping, Action-Distribution Diagnostics, Waiting Time, Explainable AI, Small Language Model, Sim-To-Real Transfer

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CHAPTER 1: INTRODUCTION

1.1 Research Background

Congestion in urban traffic is getting worse with population growth, increase in the number of vehicles owned and with the increase in the demand for travel. The signalized crossroads often turn into a bottleneck where a small error in timing of the phases can be multiplied across the network resulting in delayed network-wide, increased queuing, increased fuel usage and emissions. The recent studies on traffic signals are more focused on coordination of the intersections since single optimization might only move the congestion downstream instead of eliminating it (Hu & Li, 2024). Adaptive signal systems do exist, but they are usually critical to calibration and might not be able to sustain high performance in the case of an irregular demand, incidents and heterogeneous traffic behavior.

Reinforcement Learning (RL) has received impetus on traffic control applications since it allows controllers to train on decision policies by repeatedly interacting with the traffic world. In this context, signal decisions are modified by the agent under a response to observed states, as well as feedback goals, including the reduction of waiting time and throughput improvement (Bokade & Jin, 2025). City road systems, however, have high interaction with many concurrent crossroads; the behavior of one crossroad can affect flows and queues at neighboring crossroads making crossroads interdependent which cannot be modeled using single-agent models. This is the driving force behind Multi-Agent Reinforcement Learning (MARL) in which each intersection is viewed as a local agent learning a task as well as the contributing body to the performance of the network (Mamond et al., 2025).

Simulation environments, in which SUMO is used, offer a responsible and repeatable method of investigating delicate road networks, traffic demand patterns, and signal control logic. Significant recent developments in simulation realism, such as better traffic generation schemes, can be used to evaluate controllers that are learned under different conditions with a higher degree of credibility (Bazán-Guillén et al., 2025). Besides, there is a direction of introducing new co-simulation primes combining sensing streams and RL that show how simulation can be applied to transfer research prototypes into more realistic data-driven experimentation (Azfar et al., 2024).

1.2 Problem Statement

Traffic signal control is dynamic and multi-dimensional in its nature: system demands change with time of the day, approach turning movements change, and congestion changes cross-lane and cross-corridor. Conventional strategies are generally based on prior set plans of timing or low responsiveness, which may create an inept performance when dealing with variable traffic- seen in form of unnecessary waiting periods, increased queue time and lowered network throughput. This is made more challenging by the fact that coordination is required: although an improvement in one intersection will lead to delay reduction in that local area, the systemwide delay caused by a failure to coordinate across adjacent junctions will still result and optimization of the system will be critical (Hu & Li, 2024). This is why, in recent studies, the coordination of MARL is assessed to minimize network-level travel time and delay instead of a junction one (Mamond et al., 2025).

Even though there is a promise in MARL, there are still several practical difficulties. Non-stationarity can destabilize learning when represented by the interaction of multiple learning agents and can negatively affect performance when the relevant process occurs at an unobservable time scale or when the network becomes larger. In addition, most experiments use simplified or artificial patterns of demand; without the design of realistic scenarios, results reported may not be generalizable. Intensified realism in simulation in terms of demand is consequent to plausible assessment and reading of results (Bazán-Guillén et al., 2025).

1.3 Rationale of the Study

The rationale behind this research is that there is an increasing demand to develop adaptive signal control techniques that can adapt to the changing traffic situations compared to the relatively fixed or moderately adaptive techniques. MARL is especially applicable as it is compatible with the distributed nature of road networks, the intersections can be trained to learn localized control policies, yet at the network scale, the intersections can be coordinated to achieve network-level goals. Recent literature shows that the practical implementation of multi-agent deep RL can be explicitly aimed at achieving the performance at the corridor level by learning timing policies that take into consideration the interaction of neighboring junctions (Hu and Li, 2024). Concurrently, coordinated research platforms and standard pipelines are becoming the norm in the implementation, comparison, and evaluation of MARL-based traffic signal control, which enhances the quality of reproducibility and benchmarks (Bokade & Jin, 2025).

Another reason is that simulation scenarios need to be based on empirical characteristics of traffic to increase the realism of evaluation. Routing and demand configuration using the observed traffic indicators can help to enhance the plausibility of the inputs and outputs of the simulation (Salenek et al., 2024). Similarly, the development of realistic traffic models in SUMO also helps in building scenarios, which are more responsive to urban demand variations in the face of different situations like rush or off-peak hours (Bazán-Guillén et al., 2025).

1.4 Research Aim

The overall goal of this secondary data study is to review and synthesize chronologically a set of evidence of the multi-agent RL algorithms for autonomous traffic signal control, focusing on optimizing intersection timings, and evaluating their performance in the simulation environment such as SUMO.

1.5 Research Objectives

- RO1- Identify and classify the dominant MARL architectures (e.g. independent Q-learning, actor-critic, feudal and graph-based methods) used for traffic signal control in recent literature
- RO2: Analyze the modelling of traffic environments and intersection topologies in simulators - in particular SUMO and similar tools - and the state, action and reward modelling for each agent.
- RO3: Compare reported results of performance outcomes from MARL controllers relative to baseline methods such as fixed time, actuated and centralized deep RL controllers in terms of average waiting time and average travel time and queue length and throughput.
- RO4: Critically evaluate open issues in the deployment of MARL-based signal control, from coordination under a partial observation to scalability to large networks, transfer from the simulation to real life and robustness to demand fluctuations.

1.6 Research Questions

- RQ1 (for RO1): What MARL architectures are most adopted for traffic signal control in recent studies, and how can they be meaningfully classified (e.g., independent learners, actor-critic, feudal, graph-based coordination)?

- RQ2 (for RO2): How do studies model traffic environments and intersection topologies in SUMO (and similar simulators), and how are agent state representations, action spaces, and reward functions defined?
- RQ3 (for RO3): How does the reported performance of MARL-based controllers compare with fixed-time, actuated, and centralized deep RL baselines in terms of waiting time, travel time, queue length, and throughput?
- RQ4 (for RO4): What key limitations and open challenges remain for real-world deployment of MARL traffic signal control, particularly regarding partial observability, scalability, sim-to-real transfer, and robustness under demand variations?

1.7 Significance of the Study

The importance of this thesis lies in the fact that it provides technical and practical contributions to the field of traffic signal control. It is technically a study of MARL in a multi-intersection scenario that is implementation-oriented, and which shows how RL formulations can be compiled into practical simulation-based controllers and tested in a controlled environment. This is reinforced by the increasing access to systems that unify research in such a way that the development and benchmarking of MARL can be standardized in traffic signal control, enhancing consistency and reproducibility of experiments (Bokade & Jin, 2025).

The economic and social benefit of reducing congestion is straightforward: reducing waiting time will lead to better commuter experience, less emissions due to vehicles stuck in traffic, and an increase in the efficiency of urban mobility on a large scale. Most recently collaborative MARL methods trained in SUMO-based systems also indicate that coordinated learning has a promising role in reducing travel time in dynamic settings and minimizing delays (Mamond et al., 2025). It enhances the applicability of the results using dataset-aware scenarios, as controllers are assessed in traffic situations, which are closer to the characteristics of real distribution demand and not entirely synthetic. This will allow more plausible advice on future research and possible deployment planning such as what performance measures can provide the most insight and what operational constraints are likely to be given priority.

1.8 Research Gaps

The MARL-based traffic signal control research has various gaps in it even though there is outstanding academic interest in the subject. To begin with, most of these implementations experience learning instability in non-stationarity: as all intersection agents knowledgeably revise their policies, the environment changes with other agents, potentially decreasing convergence reliability. Second, scaling is an issue, as the complexity of city-scale networks grows, and they become more difficult to train and more highly sensitive to hyperparameters. Third, the issue of partial observability is here to stay since an intersection in most cases lacks complete knowledge of upstream and downstream conditions; and without well-developed coordination mechanisms, this will restrain the quality of decision and efficiency at the network level (Hu & Li, 2024).

One more critical void is the assessment of realism and transferability. Much of the literature is based on simplified or artificial demand patterns, which cannot be expected to give generalized results in respect to variation in actual traffic variations. Recent work on more realistic urban traffic within SUMO suggests that the significance of believable demand modeling as well as the continued desire to have more realistic scenarios in the evaluation pipelines are still needed (Bazán-Guillén et al., 2025). Besides, sim-to-real transfer and resilience to demand changes are not studied, particularly in environments where uncertainty, sensing, and heterogeneous road participants generate environments that cannot be simulated easily.

1.9 Structure of the Thesis

Chapter 1 presents the background of the research, presents the traffic signal control problem, and reasons why MARL and SUMO-based simulation offer an adequate basis to perform this study. It also outlines the research goals, objectives, research questions, significance and gaps in the research.

Chapter 2 surveys the literature regarding traffic signal control strategies, the principles of reinforcement learning, the strategies of multi-agent learning and the usual evaluations metrics and bases in this context, considering recent simulation-based strategies.

Chapter 3 describes the methodology, such as the setup of the SUMO environment, the setting of the scenarios, guided by the data, MARL formulations, and the design of the experiments.

Chapter 4 includes findings and comparative analysis.

Chapter 5 presents the important findings, limitations and future work and real-world validation recommendations.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction to Traffic Signal Control

Traffic signal control is an indispensable aspect of transportation management in the city, the work of which is to control the movement of both vehicles and pedestrians through crossings. Fixed time and actuated systems of traffic signal control have been mainstreamed since they are simple and work well in predictable conditions. Fixed-time control is defined as a system that is controlled by programmed signal cycles that do not change during the day and actuated systems that use real-time traffic flow sensors or cameras to change the signal (Rafique et al., 2024). These solutions, however, fail in most instances to be efficient even when the condition of traffic changes unpredictably, i.e. during rush hours, accidents or atypical demand patterns. It may cause too much waiting time, queues, and the inefficient flow of traffic (Miletić et al., 2022).

The main problem of traditional systems is that they cannot be changeable according to the features of the actual traffic movement. Fixed-time systems are also dynamic, implying that they cannot dynamically react to the changing traffic loads, whereas actuated systems are restricted by the ability of the sensors to correctly track the real-time information (Jayawardana et al., 2021). Therefore, conventional control techniques are not effective in handling complex and highly variable conditions in the traffic, which in most cases results in congestion, inefficiency and more emissions.

The way out of these limitations is the creation of adaptive signal control systems. Such systems dynamically change signal timings according to the present traffic conditions to provide a higher level of flexibility and efficiency when compared to a traditional approach. They also rely on traffic data as a means of constantly optimizing signal phases and cycle lengths to increase responsiveness to traffic demand. The main advantage of the adaptive control systems is that they can decrease the number of vehicles in the congestion period and then lead traffic across more intersections through optimization of their flow (Rafique et al., 2024). Nonetheless, adaptive systems may still have limitations due to limited data to work with, underlying network interactions, and the computational resources; thus, more sophisticated systems, including intelligent and learning-based systems, are necessary (Miletić et al., 2022).

This transition to intelligent systems in traffic control has inspired the start of studying the field of Reinforcement Learning (RL) and applying this technology to traffic signal control via Multi-Agent Reinforcement Learning (MARL). Such systems based on learning have massive benefits when it comes to the handling of intricate and dynamic traffic systems. RL algorithms allow agents (i.e., traffic signal controllers) to discover optimal signal timing policies during interactions with the traffic environment and increase performance because of real-time feedback. This approach is further expanded by the usage of MARL that enables multiple intersections (agents) to learn and adapt each other but act jointly to optimize their overall network performance (Satheesh & Powell, 2025).

2.2 Reinforcement Learning (RL) Fundamentals

Reinforcement Learning (RL) is a subdivision of machine learning in which an agent develops decisions because of experiential exchange with the setting and has feedback in the form of rewards or sanctions. An RL agent is aimed at maximizing cumulative rewards given by actions taken over time to achieve positive results. RPRL operates on the concept of trial and error wherein the agent tries alternative actions and perceives the outcomes of its actions, which positively changes the process of decision-making by improving it over time.

The environment is generally represented in the form of a Markov Decision Process (MDP), and in RL, there are four main elements of the MDP, which are States, Actions, Rewards and Policy. States are the prevalent state of the environment like traffic movement or congestion at a crossroad. Actions are the actions that can be taken by the agent like the time of the traffic lights. Rewards are the feedback which is given to the agent following an action that has been taken that solidifies the responses that create the desired results (e.g. reduced waiting time) and charges those actions that create the unwanted results (e.g. increased congestion). The Policy is a plan of action of the agent to decide on the further course of action depending on the status quo.

The RL agent aims to find a policy to maximize the cumulative reward of the expected value which incurs a tradeoff on exploration (researching novel actions with the aim of finding more successful ones) and exploitation (selecting the most successful action according to experience). This is the exploration-exploitation trade-off on which RL is based and guarantees that the agent does not only optimize on the known outcomes, but it also finds new, possibly more effective strategies.

Q-learning is amongst the most famous RL algorithms. Under Q learning, the agent gets trained around action-value function, denoted Q(s, a), which approximates the anticipated future reward of doing action a in states. The Q-values are iteratively updated on the feedback of the environment, and it slowly gets to an ideal policy. It has been used successfully in traffic signal control where Q-learning is employed to train agents to change signal distribution times to reduce congestion and waiting times to attain better traffic flow at the intersection.

The other RL approach is the Policy Gradient Methods that directly minimize the policy by updating the policy parameters according to policy reward gradient. The use of these methods is especially effective in problems where there are large or continuous action spaces like traffic signal control at a combination of two or more intersections as the action space is prohibitively large to manage by standard Q-learning.

Over the recent years, Deep Reinforcement Learning (DRL) has become one of the most potent methods that apply deep neural networks as a source of approximations to either a Q-function or directly the policy. DRL is specifically useful in large state and action space environments, including traffic signal control, where the more traditional RL would prove difficult to scale. DRL has already been used to solve the problem of traffic signal systems, where agents prevent a need to adapt to changing traffic conditions and reduce signal timings at several junctions in real time (Prajapati et al., 2024).

Nevertheless, there are several challenges of RL in traffic signal control. The major problem is non-stationarity, taking place when several agents update their policies at the same time because of which the learning process becomes unstable. Also, RL approaches often can demand a massive number of interactions with the environment to find optimal policies, and this may be computationally expensive and time-consuming, particularly when dealing with a large-scale traffic network. Multi-Agent Reinforcement Learning (MARL) has been proposed to manage these difficulties; that is, multiple agents (models of intersections) learn and make decisions autonomously, and coordinate them to achieve the maximum performance on a network level. Whereas MARL allows distributed learning, it retains system-wide goals (Satheesh & Powell, 2025).

Nonetheless, MARL is confronted with the challenge of maintaining stability and convergence particularly when the information that agents have about the state of other agents is limited, causing environments to be partially observable. The solution to these difficulties is the elaboration of sophisticated algorithms that provide successful communication, coordination, and scalability to implement the MARL practically in the systems of traffic lights control in the real world.

2.3 Multi-Agent Reinforcement Learning (MARL) in Traffic Signal Control

Multi-Agent Reinforcement Learning (MARL) is an extension of the classical Reinforcement Learning (RL), where various agents cooperate, learn and make decisions in a common environment. Under the framework of traffic signal control, every intersection is formulated as a free-acting agent that seeks to maximize traffic moving through the intersection by setting signal timing in respect to available real-time data. MARL as opposed to single-agent RL: In the single agent RL there is only one agent which lives in contact with the environment, in MARL, there are multiple agents which have to coordinate the decisions so that the actions of a single intersection do not hurt the actions of neighboring intersections thereby the actions of the whole network can be optimized (Michailidis et al., 2025).

2.3.1 Types of MARL Algorithms

MARL algorithms have various types, and each solves various conditions of learning and making decisions in multi-agent settings. Among the easiest methods is Independent Learners (e.g. independent Q-learning) where each agent does not pay attention to the behavior of other agents but acts as an independent learner. Although this method is less computationally focused, it may encourage diminished performance, because of the absence of coordination among agents (Satheesh & Powell, 2025).

Conversely, Cooperative MARL, e.g. Actor-Critic or Q-Learning with communication, enable the shared exchange of information and communication between agents in a bid to enhance performance. In Actor-Critic approach, an actor makes decisions regarding the action as per policy at that time whereas the critic appraises the action by estimating the value function. This allows the agents to make better-informed decisions, as they can vary their behavior owing to the response of the environment and the agents. Cooperative approaches enhance the level of coordination but necessitate extra communication infrastructure amid agents (Miletić et al., 2022).

The other difference in the MARL approaches lies in the Centralized and Decentralized approaches. The centralized MARL is characterized by a central unit that collects the information available to all the agents and makes decisions on the optimal joint action. The result is the best decision since it takes into consideration the global environmental conditions. But it is also computationally intensive, and it is not feasible in large systems. In the case of decentralized MARL, that is, individual agents are exposed to independently learning and making decisions. Although this solving method is more scalable and efficient, it can face the problem of coordination particularly with large networks that contain several intersections (Miletić et al., 2022).

Even higher-order algorithms of MARL are Feudal MARL and Graph-Based Approaches. Feudal MARL splits the issue of decision-making in sub-tasks and allocates each agent a given role. This hierarchical setting allows the agents to learn policies of higher efficiency and large-scale networks. Graph-Based MARL, in its turn, consists of picturing the environment as a graph with intersections (agents) as nodes and their interaction as edges. This method is especially applicable in the urban traffic scenario in which it becomes easier to model the interactions between the agents, and these agents can easily cooperate and exchange information to manage the traffic in an optimum way (Satheesh & Powell, 2025).

2.3.2 MARL for Traffic Control

Applying MARL in traffic signal control will optimize the timing of lights at crossroads with several crossroads for a specific scenario by employing the collaboration of various agents. The cooperation of the agents is marketable since the evaluations of a single intersection have the potential to impact the movement of the other intersections. As an example, a green intersection can be too long and thus can create congestion on the adjacent intersections and hence inefficient traffic flows. MARL algorithms make the intersections adjust dynamically to the local traffic circumstances considering the effects on the whole network.

Nevertheless, coordination among agents has several problems. Partial observability is one of the most important problems, as not all the agents know the full picture of the surrounding, including the state of traffic at other crossings. Such incomplete information may result in nonoptimal decisions and a decrease in the overall performance (Rafique et al., 2024). In addition, agents may lack the ability to exchange valuable information regarding the traffic conditions due to the communication constraint which may further complicate the decision-making process.

Despite these issues, MARL has demonstrated positive outcomes to enhance traffic signal control. An example is Bokade & Jin (2025) which showed that cooperative MARL approach was effective in controlling traffic flow at more than one intersection, indicating that the waiting time and total congestion were greatly minimized. Mamond et al. (2025) also applied MARL to optimization of signal timings in an urban network that was quite complex, and the results were better compared to the traditional fixed-time and actuated control systems.

2.3.3 Benefits and Limitations of MARL in Real-World Traffic Signal Control

The advantages of using MARL in the context of traffic signal control are also evident: it can offer adaptability, data-driven solutions that can always update the timing of signals depending on real-time traffic data. MARL systems are dynamic and can smoothly adapt signal timings to lessen congestion, waiting stages and enhance total network effectiveness, unlike the traditional systems which are normally fixed or necessitated laborious tuning.

Nevertheless, the application of MARL in the case of real-world traffic signal control is limited. To begin with, there is the issue of scalability of MARL approaches. With the increased number of crossings in a city, the system becomes increasingly complex exponentially. This may result in time-consuming training as well as efficiency in calculations. Also, multi-agent environments are generally not stable, and at any time, a multi-agent can lead to instability in the learning process (Shen, 2025). Besides, there is also the problem of sim-to-real transfer; where MARL models perform well in simulation, such as SUMO, in real-life traffic systems, implementation is frequently complicated by the dynamism of traffic, as well as other external elements, such as weather or special events (Ye et al., 2025).

2.4 Simulation Platforms for Traffic Signal Control

Traffic simulators are essential in assessing and investigating the Multi-Agent Reinforcement Learning (MARL) algorithms to control traffic lights. They offer a restrictive setting in which numerous conditions can be modeled in which a researcher can experiment with alternative methods of traffic flowing optimization without putting lives at risk, as would otherwise have been the case with real-life experimentation. One of the most popular simulation systems is SUMO (Simulation of Urban MObility). SUMO is a free, micro scale, traffic modeling simulation tool which simulates the traffic behavior and dynamics, movement of vehicles, and city signals a crossroads networks. It is very flexible in the model of complex road systems, the ability to deal with various types of traffic flows, and the ability to accommodate diverse types of vehicles and sensor configurations (Ye et al., 2025). SUMO is especially coveted in its capability to model the dynamics of large-scale networks that bear seminal traffic dynamics, which makes it the most perfect to assess the efficiency of MARL-powered traffic control measures.

Though other simulators such as VISSIM and AIMSUN are also utilized in the study of traffic signal control, SUMO is frequently used due to its open-source characteristics, scalability, and the opportunity to be combined with other external tools and algorithms. In contrast to VISSIM that is proprietary and more expensive to use, a researcher has access to a low-cost platform to build on in SUMO and tailor it to the requirements of a particular experiment. To provide an example, the fact that SUMO can be interconnected with Python-based frameworks and deep reinforcement learning (DRL) algorithms can provide an opportunity to develop MARL algorithms directly into the simulation environment and make real-time learning and adaptation possible (Kusari et al., 2022). This flexibility contributes to the fact that SUMO is considered the choice of most people in the sphere of traffic signal control research.

Simulation is critical to MARL because it can be used to offer controlled and repeatable experimentation. Traffic systems in the real world are affected by a myriad of unknown factors that are generally unpredictable, like weather, accidents, or road users, which can be very difficult in assessing the practices of traffic signals control measures. Through new simulation platforms like SUMO, the computer-based researcher can recreate a given state of the traffic, test new signal control strategies and measure the outcomes of performance measurements like waiting time, queue length and throughput among different controlled scenarios. These sim games enable one to carefully evaluate the performance of MARL algorithms in realistic, but controlled, conditions (Reza et al., 2023).

Also, traffic scenario modeling is a sought-after feature of simulation-based testing. Simulation tools that are important to capture the dynamics of a city traffic scenario include demand data, which in this case needs to be realistic in terms of traffic volumes and vehicle distributions. Such data may be obtained by real-life traffic investigations or be produced by statistical models replicating the traffic patterns such as peak or off-peak situation. Simulations with real-world traffic data can help the researcher to be certain that their MARL-based algorithms are evaluated in the context, which most closely reflects the reality of actual traffic conditions and guarantees that the achieved results have better validity and applicability (Rafique et al., 2024).

2.5 Reinforcement Learning Applications in Traffic Signal Control

2.5.1 Historical Context: Early Applications of RL in Traffic Signal Control

The use of Reinforcement Learning (RL) in traffic signal control started as a means of optimization of the traditional fixed-time systems. The initial RL-based techniques aimed at optimizing the timing of signals at isolated intersections through the optimization of signal stages according to real-time traffic states to reduce the waiting time and congestion (Miletić et al., 2022). Single-agent RL algorithms mostly formed the basis of these early applications where there was a single intersection, which served as an agent, making decisions under local traffic observations. Although it worked well in small, isolated intersections, it was challenging to scale large and interconnected traffic networks.

2.5.2 Single-Agent RL for Traffic Control

In single-agent RL, the agent is trained to tune the time of a signal, based on the traffic conditions (including the number of vehicles and waiting time). One of the most popular RL algorithms, Q learning, is an algorithm that approximates the future rewards of executing actions in a particular state, which guides the agent through the right signal phases (Rafique et al., 2024). This approach proved useful on relative predictable traffic patterns in isolated junctions. Nonetheless, it does not perform well in many intersections as the behavior of one intersection influences other intersections causing poor performance without coordination between the interfaces.

2.5.3 MARL for Multi-Junction Coordination

Multi-agent reinforcement learning (MARL) was proposed to overcome the single agent RL limitations. MARL considers intersections as individual agents that learn to maximize signal times depending on the local traffic and requires coordination with other agents to maximize the networkwide performance. MARL permits intersections to engage and coordinate their behaviors to enhance the general traffic movement in the network (Satheesh & Powell, 2025). MARL has been effectively implemented to multi-junction optimization to facilitate a decreased waiting time and high throughput and surpassed traditional fixed-time systems (Miletić et al., 2022).

2.5.4 Successes and Limitations: Case Studies of MARL Applications

Some studies have demonstrated that MARL can be successful in traffic signal control. Indicatively, Mamond et al. (2025) showed a collaborative MARL process to signal optimization in various cross-intersections to minimize congestion and waiting times. Hu & Li (2024) use deep reinforcement learning (DRL) in the case of multi-agent systems and demonstrate a better traffic flow in comparison to traditional ones. Although these have been successful, such issues as non-stationarity (when the agents fail to distinguish between the learning process of their opponents) and partial observability (when the agents do not have full information about the network) are still present.

2.5.5 Comparison of Traditional Methods

MARL has clear benefits compared to fixed-time and actuated control systems. Whereas the fixed-time systems operate on set schedules, and thus inefficient, and the actuated control changes with incomplete sensor data, MARL changes on the fly, as the traffic conditions in real-time change. Such flexibility enables MARL-based systems to have a better capacity to manage congestion and waiting times in more intersections compared to the traditional approaches in dynamic environments (Prajapati et al., 2024). Nevertheless, the optimization of MARL in terms of scalability and the associated computational complexity is also an issue, especially in large city networks, and it needs to be improved through future studies in terms of efficiency and performance in practice (Miletić et al., 2022).

2.6 Traffic Demand Models and Data Sources

Traffic demand modeling can be instrumental in training MARL agents to control traffic signals by supplying them with real traffic conditions on which to simulate. When traffic flow is properly modelled, then it is possible to make reinforcement learning algorithms capable of responding to real traffic patterns to modify the timing of the signals. Traffic flow is commonly modelled either in macroscopic or microscopic models which describe traffic at different levels of abstraction. Macroscopic models are based on aggregate values, like flow, density, and speed in the whole network, and microscopic models predict individual vehicle behavior and interaction at a certain intersection (Miletić et al., 2022). SUMO (Simulation of Urban MObility) is a microscopic, detailed, and low-level model, which is capable of simulation of fine-grained traffic behaviors within a road net, such as vehicle paths, crossing, and light timing (Ye et al., 2025).

Traffic data in the real world is essential in training MARL agents because it makes sure that the simulations match real-life set-ups and not artificial principles. The sensors, cameras and other tracking technologies installed in urban areas tend to gather data on the number of vehicles, the average speed, occupancy, flow rate etc. This information can be further incorporated into the traffic simulators such as SUMO to generate the situations which match the ones in the real traffic (Rafique et al., 2024). With real-world traffic data, it is possible to measure how well MARL controllers perform in conditions that represent variability and complexity of the actual urban traffic conditions.

Data on traffic, including that of time-of-day models and seasonal demand variations are important in producing natural scenarios of traffic in simulation research. Such datasets are useful in modeling the changes in the traffic movements at various times of the day (e.g., rush hours) and days of the week, which have a great influence on the scheduling of signal control. An example is a time of the day model that may give rise to increased vehicles and reduced speeds during peak hours, which would require alternative signal approaches than during off-peak periods (Satheesh, Powell, 2025). The availability of such data can be used to produce the patterns of traffic that reflect the real-life situation better and enhance the reality and practicality of the outcomes of the simulation.

Complexity of real-world variability is one of the greatest problems during the models of traffic demand. The traffic conditions are subject to very many factors including weather, special events, accidents, roadworks, which are not easy to predict and include them in the standard models. Moreover, to have traffic simulators precisely representing these real-world variations, a lot of data needs to be gathered and adduced. Consequently, the traffic signal controllers generated using MARL should be able to respond to such changing conditions and guarantee that the solutions obtained in simulations can be successfully implemented in practice (Prajapati et al., 2024).

2.7 Evaluation Metrics for Traffic Signal Control

To determine the effectiveness of the traffic signal control strategies, several performance measures are used to measure the extent to which these systems keep the traffic flowing positive as well as reduction in congestion. Average waiting time is one of the most used measures because it is concerned with the amount of time spent waiting by vehicles in the intersections. The direct positive effect of a decrease in waiting times is an increase in traffic flow, a decrease in consumption of fuel-related emissions, and a reduction in emissions (Miletić et al., 2022). Reinforcement learning (RL) algorithms, especially MARL, can be used to optimize the signal timings to reduce intersection delays by adjusting signal phases due to real-time traffic conditions.

The next valuable measure is the queue length which is used to determine the number of vehicles queuing at an intersection. When there are long queues, it is usually an indication of inefficient signal control, where a traffic jam is caused by ineffective timing, or blockage in soonest intersections (Satheesh and Powell, 2025). One of the desired actions of any traffic signal control measure is the reduction of the queue length and the traditional MARL-based systems have demonstrated the potential in governing the queue length effectively by adjusting to demand variations and minimizing the chances of bottlenecks.

Another important performance measure is throughput, the number of vehicles that were served across the intersection during a given time. Throughput maximization will guarantee that there will be proper accommodation of traffic flow through the system without excessive delays. It is especially significant in the urban world where there is a lot of traffic. This proves useful because MARL controllers can also improve throughput by dynamic changing signal schedules as more recent traffic changes would have each signal phase optimized to the current conditions, resulting in a smoother flow of vehicles over the network (Miletić et al., 2022).

Another valuable measure to assess the efficiency of the use of traffic signal control is travel time, or the average time of the vehicles that will pass through the whole network. Traffic signal controllers can help get a vehicle traveling along the entire road network through the state of less waiting at individual intersections, as well as increase the efficiency and speed of the commute (Rafique et al., 2024). This measure is especially beneficial in measuring the work of multi-intersection control systems, where the overall optimization of the network is required to be done.

These metrics are usually incorporated into traffic signal control simulation experiments to compare the performance of MARL-based systems with the traditional control systems. Fixed time and actuated control have traditional approaches that are not dynamic to the real time states of the traffic. In comparison, controllers using MARL can constantly learn and change signal timing depending on the observed traffic situation, and it provides a solution that is dynamic and responsive. These strategies are usually tested and compared in the framework of simulation platforms, such as SUMO, where researchers can measure and compare the performance indicators, waiting time, queue length, throughput and travel time, in a controlled environment (Ye et al., 2025).

There are, however, shortcomings of applying these performance metrics. Although these measures are useful in the context of efficiency of traffic lights control, they might not be the ultimate measure in the complexity of the actual traffic dynamics. As an illustration, the safety of traffic operations, e.g., the risk of accidents at intersections because of unexpected changes of signals, is usually not covered by the traditional metrics. Also, the issue of fairness, i.e., making sure that all road users (i.e., vehicles, pedestrians, cyclists, etc.) are considered during the control of the signalizing means) can be regarded as another relevant indicator that cannot be represented completely by conventional measures (Prajapati et al., 2024). Furthermore, the permanency of the traffic system is another important parameter that should be taken into consideration not only in dynamic traffic systems but also in unstable conditions that can occur because of accidents and weather conditions.

2.8 Challenges and Open Issues in MARL for Traffic Signal Control

Multi-Agent Reinforcement Learning (MARL) as an approach to traffic signal control has a variety of challenges and open research questions, which must be addressed to be successfully implemented in a real-world scenario. Non-stationarity is one of the most important problems of MARL on traffic signal control. The environment is now dynamic and unpredictable in the setting of multi-agent systems, where multiple agents (intersections) are learning and updating their policies at the same time. The inability of MARL systems to reach a steady state may lead to instability and result in the result that agents do not reach an optimal solution (Satheesh & Powell, 2025). This poses a problem since the behavior of a given agent would affect the state of other agents making the learning process difficult.

The other challenge is partial observability. In a real-world traffic network, every intersection agent usually only knows a little about the entire traffic situation throughout the entire network. Since the agent is not in full information, the agent might not be able to make any informed decisions and therefore his performance may be suboptimal. To illustrate, when an intersection does not realize what the state of traffic is on a nearby intersection, it can make decisions that will worsen the conditions in other areas of the network (Miletić et al., 2022). To cope with this face, there have been suggestions for employing strategies that achieve communication between agents and centralized learning, whereby information is shared by agents to enhance their decision-making processes.

Another major concern in the implementation of MARL in a large city network is scalability. The size of a city determines the extent of coordination of traffic signal control within the network as the intersections are many. Having many agents in a large system would consume a lot of computational power to train, and result in increased training time. What makes this scalability problem more complex is the fact that there is a need to keep the agents in touch and make the system work in real-time. The studies are continuously underway to create more effective learning algorithms and architecture, which can be scaled to bigger networks and make it perform (Rafique et al., 2024).

Another obstacle towards MARL in traffic signal control is transferability. Algorithms of MARL are also testable and variable in simulation; however, extrapolation of the simulations in real-life contexts is a major challenge. MARL-based systems are not always as capable of performing within the context of actual traffic systems as they do within a simulation due to limitations of the sensors, the variables of the environment being unmodeled, and due to real-world traffic variations (Jayawardana et al., 2021). It is important to guarantee that the MARL controllers can be successfully implemented in the real urban traffic networks by ensuring they are able to transition between a simulation environment and a deployment environment.

Lastly, resistance to change of demand is a very important issue facing MARL-based traffic signal control systems. Demand may change dramatically during the day because of accidents, road closure or special events. The MARL controllers should have the capacity to ensure that the changes are adjusted within a short period of time to sustain the optimal traffic flow. It involves the necessity to have fast-learning algorithms capable of modifying the signal timings based on real-time information and stabilization them, allowing them to remain stable and efficient even in known traffic unpredictability (Miletić et al., 2022).

CHAPTER 3: METHODOLOGY

3.1 Introduction

The study is designed to explore the capabilities of the field of multi-agent reinforcement learning (MARL) to optimize traffic signal control. The research aims at the strategy of optimizing the interview duration and alleviating traffic congestion in the city through the application of MARL methods. Using secondary data, and simulation environment, including SUMO, the study attempts to analyze the extent to which these algorithms can effectively be used in practice, with reference to the actual traffic situations. SUMO makes traffic network simulation a flexible and open-source platform, which is an indispensable resource when evaluating the MARL-based traffic signal controllers (Michailidis et al., 2025).

In this respect, real-world traffic systems secondary data are important. The data utilized in this research comprises key traffic related measures like the number of vehicles, mean speed, occupancy of the lanes, rate of flow, and waiting time. These variables will show the dynamics of the traffic flow and create the basis to evaluate the work of the MARL-based controllers. Past research data have shown that real-life data results in better simulation of traffic environments to facilitate a refined model training and evaluation (Xiao et al., 2024). This data is to supplement the study because it provides realistic traffic data that will be involved in the evaluation of MARL controllers performance in the management of intersections controlling timings.

3.2 Secondary Data Review Approach

The research design applied in this study is found on secondary data review, which is a systematic examination and synthesis of empirical research based on peer-reviewed articles, conference papers, and technical reports about MARL in traffic signal control. This method is selected as it permits us to combine a large set of insights and conclusions of the available literature without the necessity to collect primary data. The analysis will allow the study to concentrate on the synthesis of the best practices and strategies that have been applied in the field before (Kolat et al., 2023).

The data used in this study contains some vital variables of traffic which are the number of vehicles on a particular road, the mean speed, the occupancy of lanes, the level of traffic flow, and the waiting period. These metrics are essential to the simulation of traffic behavior as well as measure the efficiency of traffic signal control techniques. This dataset was chosen to fill the current volume of research concerning the importance of optimization of traffic signals because it offers desirable real-world information, which is often deficient in simulation-based studies. Its inclusion provides stronger credibility to the research since the presented information is based on real-life practical and observed conditions in traffic, which are necessary to evaluate the feasibility of the MARL algorithms (Mushtaq et al., 2023).

In this study, the primary data collection is not involved, and the secondary analysis of the existing data and literature will be used. The secondary data analysis also makes it possible to investigate more efficiently since it will use the information that has already been obtained, which means that the study will be aimed at analyzing and synthesis of the most essential findings. The literature that exists offers a holistic view of the current situation with the MARL-based traffic signal control with much of the information being obtained through simulation-based experiments that utilize software such as SUMO (Goel et al., 2023). The simulations prove to be essential in testing the MARL algorithms because it is possible to test the different traffic situations without risks and expenditure of conducting the experiments in the real world.

Moreover, the review of secondary data will provide an opportunity to conduct a general overview of different strategies of traffic signal control and their results. The collected data will be compared with other comparable works to come to conclusions and conclusions about the performance of MARL in traffic control (Swapno et al., 2024). As an example, the effectiveness of MARL controllers will be investigated in terms of such metrics as average waiting time, throughput, and traffic flow, and the results will be compared to the traditional ones, including fixed-time and actuated control systems (Louw et al., 2022).

This methodology is effective in terms of best practices within the discipline and is aimed at providing an assurance that the findings of the study may be applied to the real life situations. This creates another gap in the existing literature, as the specific strategy will help to generate the synthesis of available data and present new perspectives on the use of MARL to optimize traffic signals.

3.3 Data Extraction and Preprocessing

There are several steps that are essential in the process of extracting and preparing the dataset to facilitate the analysis process so that the data is clean and prepared to be modeled. To begin with, the raw data should be isolated out of the traffic data so that the traffic information contained different variables, which include the number of vehicles, an average speed, lanes congestion, speed and flow rate, and the waiting time. The preprocessing involves the first step in processing any missing values. In case of data loss, suitable measures are taken to cover such spots, like imputation (substituting the absence of data with the average or the median), so that the data will not have any gaps that might bias the analysis (Michailidis et al., 2025).

The normalization or scaling may be required after the treatment of the missing values to achieve uniformity among variables. Some features, including flow rate and number of vehicles, could be of varying scale, and it is important to standardize them. This will make all variables equally contribute to the analysis and the models will not be skewed towards those variables that have bigger numerical ranges (Xiao et al., 2024).

Other time variables like the time_of_the_day are also converted while preprocessing. Because time may influence traffic and the time waiting, it is transformed into either one of the following categories (mornings, afternoons, evenings) or numeric variables (hour of the day) to be represented in the analysis. Such changes make sure that the time-related trends can be appropriately reflected in the model (Mushtaq et al., 2023).

The most significant characteristics which are relevant in the analysis are the number of vehicles, flow rate, and waiting time. The intensity and level of congestion of traffic will be measured based on vehicle count and flow rate, whereas wait time is considered one of the performance measures to optimize the timing of traffic signals (Kolat et al., 2023). These attributes, after up-processing, become the basis of the information in the dataset to gauge the effectiveness of MARL controllers in recreating the dynamics of real-life traffic scenario.

3.4 Simulation Environments and MARL Architectures

Evaluation of the performance of MARL-based traffic signal controllers requires the involvement of simulation platforms. An open-source and adaptable simulator is called SUMO (Simulation of Urban Mobility) which is one of the most popular platforms. SUMO can be used to model traffic behavior at both the micro and macro level, and this makes it suitable in simulating traffic flows and intersections in various scenarios (Louw et al., 2022). The feature of SUMO to model realistic traffic flows based on empirical measures of urban settings mean that MARL algorithms are trained in an environment that is close to realistic traffic (Goel et al., 2023).

In the literature, several MARL architectures have been found, which propose various approaches to traffic signal control optimization. One of the first and most general algorithms is independent Q-learning in which agents (intersection) learn autonomously using their own experience. Nevertheless, the strategy might not work with convergence and performance up to the point of more complex network in larger cities (Swapno et al., 2024).

A more advanced approach is actor-critic frameworks, in which an actor takes a decision about the further steps based on the policy where a critic assesses the performance of that decision estimating an action by value of the decision chosen. The MARL architecture is also a more efficient form of the agent because it enables agents to update their strategies, regarding their experience as well as their agent responses (Xiao et al., 2024).

Graph-based coordination has been of recent interest as a viable technique towards enhancing coordination between junctions of large networks. Such approaches constitute graph theory models in which intersections are represented as nodes and interactions between them as the edges that provide the system with dynamically changing response to changes in traffic patterns throughout the network. These architectures enable an interactive control of the traffic signals by including, into the learning process, the intersection of the neighboring intersections data about the traffic (Kolat et al., 2023).

The dataset is highly critical in the training of MARL agents via simulated realistic traffic environments. The characteristics of traffic intensity and congestion at every intersection are modeled through features like vehicle counts and flow at every intersection. As an example, when the number of vehicles is high, the MARL agent can modify signal timings so that the number of vehicles entering the traffic increases (Swapno et al., 2024). Also, occupancy and average speed of lanes are taken to determine the condition of the traffic and make a better decision regarding the signal phases. When the lane occupancy is high, and the average speed is low, this can be an indicator of congestion, and thus the algorithm can change the timing of the signals to reduce traffic congestion (Ammari et al., 2025).

Waiting time , a major characteristic in the dataset is adopted as a reward or a cost measure in the reinforcement learning process. The MARL agents are aimed at reducing waiting time for vehicles thereby maximizing the traffic flow. In that regard, the waiting time will be the most relevant performance indicator that will direct the learning process because the agents will strive to decrease delays without impacting the effective movement of traffic through several intersections (Mushtaq et al., 2023). This reward design would play a valuable role in making sure that the congestion is alleviated and efficiency is enhanced by the MARL-based controllers.

Such MARL implementations and simulation environments, which are coupled with real traffic data, offer the basis of testing and optimizing traffic signal control policies with MARL. Using both the theoretical knowledge of the MARL and the practical information of the urban traffic simulations, this study will suggest the solution to the issue of improved coordination and optimization of the urban traffic system.

To validate the theoretical framework in a live simulation setting, a SUMO-based experimental environment was additionally implemented. A SumoTrafficEnvironment class was developed to wrap the TraCI API into an OpenAI Gym-style interface, enabling the same Q-Learning and DQN agents evaluated on the dataset to interact directly with a running SUMO simulation without modification. A single 4-way intersection network was programmatically generated with stochastic vehicle flows across all 12 turning movements. The state space comprised seven features observable via TraCI — queue lengths on North-South and East-West approaches, total vehicle count, mean speeds per approach, current signal phase, and remaining phase duration. The action space retained the same three-action design: decrease, maintain, or increase green time by five seconds. The reward function penalised total queue length and mean waiting time per step.

3.5 Performance Metrics and Evaluation Criteria

The performance measures are essential in determining the efficiency of MARL-based traffic signal controllers. The main metrics considered in this study to evaluate were the average waiting time, the queue length, the throughput, and the average travel time. These measures give us details about the effectiveness of the traffic signal control system because they help understand the extent to which the system minimizes the congestion issues and maximizes traffic movement. One more relevant indicator is average waiting time because it has a direct effect on the level of comfort and performance of commuters and a decrease in it enhances the overall traffic experience (Michailidis et al., 2025).

These measures are measured based on the dataset that is given to analyze them. Based on the data of arrival and departure of vehicles, there is a calculation of average waiting time and line length and this includes time vehicles spend waiting at the crossroad. The measurement of throughput is achieved by calculating the frequency of the vehicles going through intersections at various points in time, and the average traveling time is measured by monitoring the total amount of time taken by the vehicles to move in the system. Such measures can describe the efficiency of the functioning of a system, and the shorter waiting time and shorter queues denote the better level of efficiency (Xiao et al., 2024).

These metrics are contrasted with the traditional signal control methods, including those based on fixed time and actuated control to benchmark the MARL-based controllers. Fixed time systems use a set of pre-programmed schedules whereas the actuated systems vary the signals using real-time traffic conditions. Nevertheless, both approaches do not seem to respond effectively to changing patterns of traffic. The controllers using a MARL, in turn, constantly learn and adjust according to the situation on the road, which results in better traffic flow and the minimization of delays in contrast to the traditional system (Swapno et al., 2024).

Waiting time variable of the dataset is also of significant interest because it will be employed as the major reward in the process of optimization in the MARL. The training of the agents aims to decrease the waiting time, and the provision matches with the aim of the study to optimize the timing at intersections and decrease delays.

3.6 Data Analysis and MARL Model Evaluation

Analysis of data is performed using the techniques of statistical and machine learning to assess the effectiveness of the MARL-based traffic signal control system. To begin with, correlations between the most important traffic measures, i.e. the flow rate, the time of waiting, and the number of vehicles, are investigated to examine all these elements in interaction and the way they affect the efficiency of the whole network. To establish the presence of any significant relationship, statistical tools like correlation coefficients and regression analysis will be employed to give information on the effect that alteration in traffic flow or waiting time has on the performance of the control system (Goel et al., 2023).

The given dataset can be utilized to reproduce various traffic conditions and to train MARL-ready agents. In these simulations, one can test different algorithms in the presence of realistic traffic. An example of such variables is vehicle count and flow rate, which can be used to simulate the conditions of increased and decreased traffic, and the occupancy of the lanes and the average speed of the vehicles that models the congestion and traffic situation at the intersection. Those aspects shape decision-making of the MARL agents when it enables them to enhance the timing of signal according to the perceived circumstances (Mushtaq et al., 2023).

In assessing the performance of these models, the controllers built using MARL are compared to the baseline approaches like fixed-time and actuated systems. Fixed-time systems operate on fixed signal timing settings that do not vary in real time with traffic conditions whereas actuated systems operate on frequency of traffic sensors to vary signal timing, though might have problems with the unpredictable traffic patterns. In comparison, MARL-based systems also dynamically manage the timing of the signal by learning on a continual basis on the current flow of traffic and achieving optimal values like waiting time, queue length, and throughput (Louw et al., 2022).

The effectiveness of the MARL models is determined by the capacity to reduce the waiting time and queue size and increase throughput and mean travel time. Such performance comparisons will help to emphasize the benefits of MARL compared to conventional approaches and show how the urban traffic can be managed using machine learning (Ammari et al., 2025).

3.7 Challenges in MARL Implementation

Multi-agent reinforcement learning (MARL) or applying the technique to traffic signal control is associated with a range of issues that are inherent to it and have been widely debated in the literature. Non-stationarity is one of the greatest challenges as the environment keeps changing as every agent (intersection) changes its policies. In traffic systems, the effect of one intersection to learn and adapt means affecting the states of the other adjacent intersections therefore causing instability in the process of learning. It is especially difficult to address in big networks because, in this case, interactions between agents are more complicated and unpredictable (Michailidis et al., 2025).

The other issue is partial observability that is the fact that not full information is provided to every agent. A real-world traffic network In a real world traffic network, interface (agent) can only access immediate traffic data, not the data of what is happening in other interfaces. The narrow perspective can also result in the poor decision making as the agents do not consider the state of all the network. Solutions such as agent-agent communication or centralized learning have been suggested to deal with the issue, yet it is still difficult to make practical solutions work in large networks (Kolat et al., 2023).

Some of these challenges will be pointed out by the dataset utilized in this study. The dataset shall enable us to determine how responsive the system is to changing traffic conditions since it simulates various traffic situations such as rush hour and off-peak. Attributes such as vehicle numbers and flow rate can give information about the suitability of the model to changing traffic conditions, particularly during the most changeable times when traffic conditions become less predictable (Xiao et al., 2024). The dataset will play a key role in determining the robustness of the MARL controllers in different situations as affected by traffic and their performance in the situations involving partial observability and non-stationarity.

In addition, the other issue that should be considered when using MARL in a large-scale traffic network is scalability. The complexity in dealing with more intersections in the training of multiple agents increase exponentially, and it is hard to scale the controllers based on MARL to city-wide networks. The scalability of MARL algorithms is also tested on the simulation environment, such as SUMO, which develops networks of different sizes and traffic conditions (Mushtaq et al., 2023).

Real life data such as that applied in this report is essential in solving the problem of simulation to reality transfer. The dataset also makes sure that the simulations are made on the real traffic data, thus the probability that the MARL controllers will be able to work in real-life urban settings is high. The practicality of the underlying simulation information serves to counter the risks of generalizing the simulation models on the corresponding real traffic networks, where other phenomena like weather and accidents could make decision-making even more complicated (Goel et al., 2023).

CHAPTER 4: FINDINGS AND COMPARATIVE ANALYSIS

4.1 Introduction

The chapter presented and discussed the empirical results of three reinforcement learning (RL) controllers in a traffic management simulator: tabular Q-Learning, Deep Q-Networks (DQN) and Q-Learning with a Small Language Model (Q-Learning + SLM). The analysis, instead of reporting final scores, aimed to show how the design decisions made by each model influenced learning processes, convergence behavior and amounted to practical control choices at training and evaluation time scales as they should in evaluation-oriented reporting of simulation-based fields in RL traffic studies (Michailidis, 2025). The evaluation of performance was performed using (i) mean episode reward and (ii) mean waiting time, since indicators of waiting-time are used as the key proxy of traffic control quality in similar simulations (Xiao, 2024). The variability of the reward (standard deviation of the reward during the final training window) was considered as a training stability metric, and the policy behavior analysis based on the distribution of actions was conducted to identify the presence of state-helpful control or the presence of degenerate dominance of the single-action policy. In case of Q-Learning + SLM, one more lens was applied: consultation behavior, such as the frequency of the override, the distribution of confidence, and the interpretability output that the SLM gives when asked at a particular point in a consultation.

The third model was an extension of the previous two-model analogy of great significance. The SLM was not a replacement of the temporal-difference updates, rather it was a periodic domain-knowledge expert which was activated at specific intervals. This mixed scheme indicated the general labor involving language-based reasoning and statistical learning to become more transparent, and still not fail to inject the efficiency of tabular RL (Hu et al., 2025). Notably, the SLM did not start at each step but after every ten environment steps. This time-bounded architecture constrained the processing workload and maintained the exploration behavior needed to fill in its value table with a variety of state-action approximations.

4.2 Experimental Setup and Measurement Protocol

4.2.1 Environment and Control Objective

The experiments were made within a traffic control setting according to the conventional RL interaction cycle. At the beginning of every episode the agent makes the call reset () to get an initial state and thereupon executes the choices sequentially selecting an action and calling step (

action) until it terminates. The steps delivered the preceding state as a scalar reward, a termination flag, and an information dictionary comprising indicators regarding the system level. Such episodical framing facilitated the possibility of repeating the policy rollouts and also maintained temporal development of a traffic state (Kusari, 2022). The control aim was to acquire a policy which minimized congestion-related delay without sacrificing feasible flow through the controlled section using a discrete action space which was used to control flow conditions.

4.2.2 State Space, Action Space, and Reward Signal

The agent noticed a compressed state representation based on traffic indicators (e.g., vehicle number, speed, occupancy of a lane, traffic flow, and waiting time). Manipulation was discrete and was treated as Decrease Flow, Maintain or Increase Flow. Reward role was designed so that it penalized delay and congestion and gave action- sensitive feedback based on thresholds (e.g. median/percentile vehicle-count and flow-rate). Consequently, the reward values were shouldered with the outcomes of traffic as well as the manner through which the shaping term reacted to specific actions within the particular state regimes. This was important to interpretation in the future: models might gain dissimilar rewards whilst generating identical waiting-time results when providing the shaping term to reward action decisions but fail to translate into operational enhancement.

4.2.3 Training vs Evaluation Horizon

Two horizons were utilized: a capped training horizon of Q-Learning with modification and uncapped evaluation horizon of all models. To ensure stability and computational comparability, there were a few times where training sessions were bounded, but when evaluating rollouts where practical bounded by the natural terminal condition of the environment. This introduced an inherent discrepancy between indicators of training curves and indicators of evaluation waiting times, an aspect again taken up in Section 4.7.

4.2.4 SLM Architecture and Integration Protocol

The SLM portion of Q-Learning + SLM was implemented as a lightweight rule-based advisor which generated structured output, namely, a situation label, natural-language explanation, an action recommendation, and a confidence score. Consulted after every ten steps through the environment. During each consultation, the read continuous traffic indicators of SLM compared them with thresholds calculated as percentiles using the training data. There were four signals to be monitored (high vehicle count, low speed, have occupancy, low flow) and the combination of active signals to be made on the recommended action and confidence. At a specific threshold (0.80), the SLM was doing a hard override of the action selection of Q-Learning when the confidence had reached or surpassed that specific threshold. Otherwise, the probabilistic combination of the recommendation of the SLM with the greedy selection of the Q-table allowed the Q-Learning to maintain the primary control of most of the states but with some periodic guidance on its actions. Their combination implied that even in cases where the SLM was run over by the greedy action, the resultant transition was nevertheless utilized to revise the Q-table and therefore the influence of the advisor might be recaptured in a statistical bundle of tabular value estimates with time.

4.2.5 Evaluation Procedure and Reporting Metrics

Each of the agents was trained on 250 episodes. The training curves to calculate reporting averaged over 30 episodes to smooth the step-level noise and bring out convergence trends. The standard deviation of the magnitudes of the raw episode reward at the last 100 training episodes was used to measure stability since the variability at the end of the window indicates that learning has entered a stable regime or is still waving. Assessment of the fixed policies (no exploration) was done in ten complete episodes of each model. The mean episode reward and the mean waiting time and the standard deviations were calculated amongst these rollouts. This identification was done by counting action choices between evaluation states and transforming them into distributions, effectively allowing the identification of performance of authentic state-responsive control, and single-action collapse at the same time. In the case of Q-Learning + SLM, the logs of consultations as well were analyzed: frequency of consultations override, and its scores on confidence were summarized and an example of explanations made by the SLM was examined to ensure that the natural-language justification provided by it did not conflict with the underlying threshold reasoning.

4.3 Baseline Behavior Prior to Changes

4.3.1 Q-Learning Baseline Performance and Variability

Prior to step capping, Q-Learning generated the large-magnitude negative training returns due to accumulated reward in the step-by-step long uncapped-episode. The trainers reward was about -290000, and it is on a scale that implies the penalty of delay fitted in numerous steps and not a clear indicator of bad quality policy (Miletić, 2022). The Q-table showed a slow-increasing reward curve, which suggests that the value was continuously getting better, but training was still non-steady: the standard deviation of the final-window in the last 100 episodes was around 2122.14. there was a fairly balanced amount of baseline action choice (in the three actions), indicating about 30-37 percent each, indicating that the policy had not fallen into one repeated control choice.

4.3.2 DQN Baseline Performance, Stability, and Action Collapse

The baseline DQN obtained an evaluation episode reward of about -294,632 and an average waiting time of about 29.46 seconds, which put it near Q-Learning with traffic performance although with another learning mechanism. The final-window reward standard deviation of DQN at baseline was about 107.05 which is significantly lower than that of Q-Learning as stabilization occurred due to experience replay and target network. Nevertheless, there was early action collapse in the DQN policy already, which chose one of the key actions most of the evaluation states. This behavior encouraged exploration-schedule adjustments (Section 4.5) and had context later action-distribution consonance.

The fact that both baseline models had similar waiting times implied that the approximation of functions alone could not be part of ensuring improved traffic results in this environment. Things have been otherwise in benchmarking work In small or effectively low-entropy state spaces in which reward is ransomed by a single aggregate measure, similar convergence has been reported (Louw et al., 2022). The same baseline observations inspired the addition of Q-Learning + SLM: since both baseline agents were not interpretable and exhibited dominance in fully single-action situations, it is uncertain whether periodic advice based on language could be used to diversify behavior and still provide justifications that were human-readable, without replacing the underlying rule of the RL update process.

4.4 Impact of Step Cap on Q-Learning

4.4.1 Motivation for Step Capping

Step capping limited the length of episode to minimize the variability length caused by longer stochastic rollouts. In traffics, long spurts enhance randomness of arrivals, queueing and recovery and enhance noisiness of rewards and delays convergence to stable estimates (Louw, 2022). Maximum number of steps per episode was set equal to 200 to ensure that training episodes were similar in duration, there were no extreme reward outliers that damaged the interpretability of learning curves, and the policy updates in the unit time occurred more often.

4.4.2 Training Reward Magnitude: Why It Changed Dramatically

Once step cap was reached Q-Learning training rewards changed to approximately -290,000 to approximately -5,700. This was a mathematical consequence of adding fewer terms of accumulation of per-step rewards and in itself was not indicative of improved or worse control. Delay-based reward shaping grows with the length of the episode; hence, smaller episodes inherently bring a smaller magnitude of returns without having to consider the quality of policy (Prajapati, 2024). Significantly, the rollouts of the assessment persisted up to the end and thus kept the scale of rewards used to ensure that step capping did not alter the measurement of training but the definition of the outcome underlying the environment in the first place.

4.4.3 Stability Improvement in Training Dynamics

The major substantive impact of step capping was a drastic increase in the stability of training. Q-Learning training rewards SD during the last 100 episodes dropped to 77.83 with step capping (almost 96 percent of the baseline, 2122.14). The cap also reduced the influence of high source variance on the consumption of systematic higher-long exposure sequences in which early choices can have vast cumulative sanctions (Saiki, 2023).

4.4.4 Action Selection Shift Under Capped Horizon

The preferred action learned by Q-Learning was also redirected by step capping. Capped training had the action distribution of Evaluation concentrated into Decrease Flow (about 89.8 percent), Maintain (4.8 percent), and Increase Flow (5.4 percent). This was a shift towards a locally efficient, short-horizon strategy which solidly minimizes immediate penalties compared to the richer balance baseline distribution. The shorter the horizon, the less the agent is exposed to the delayed effects of congestion build-up and recovery, and therefore the biasing effect of TD-updates to take actions that minimize short-term penalty despite being myopic in terms of throughput (Rafique, 2024). This directional preference was later used as a guide in explaining the advisory bias to the SLM and how it was introduced into Q-Learning + SLM.

4.5 Impact of Exploration Schedule Changes on DQN

4.5.1 Why Adjust Exploration in DQN

DQN relied on epsilon-greedy exploration to come up with varied transitions to feed its replay buffer. Epsilon decay was also too fast; this means that the agent would commit to an action prematurely and the replay buffer would look behaviorally homogeneous. Instead of learning state discrimination, the network would then strengthen a stringent policy. It was a very acute risk in the context of traffic control, in which non-linear queue dynamics and demand that vary over time may necessitate long-term exploration to find robust policies (Prajapati, 2024). To allow meaningful exploration and slow down a catastrophic breakdown, the epsilon_decay was set to 0.995 and epsilon min was set to 0.01.

4.5.2 Effect on Evaluation Performance and Persistent Policy Collapse

Following the change in the exploration schedule, the mean evaluation reward of DQN was slightly worse than it was with the previous exploration schedule, changing by about -294,632 to -295,086.92 (a marginal degradation), but the wait-time was fairly constant at about 29.46 seconds. Such findings suggested that the refined exploration parameters could not reveal a qualitatively better traffic policy that could be found within the training budget. What is more important, action distribution analysis revealed that policy collapse prevailed: DQN chose Increase Flow on 100 percent of the sampled evaluation states. The persistence that collapse was not necessarily due to the exploration speed but probably by the reward structure or state representation, which can have lacked sufficient discriminant signal such that alternative action can have become consistently more preferable (Bouktif, 2023). Variability in training also changed relatively little (standard deviation decreased by 107.05 to 85.07), and the diagnostics showed that the training quickly reached a high replay buffer and the loss plateaued in a small-scale oscillation indicative of transition into a fixed-point behavioral regime.

4.6 Cross-Model Comparison After Changes

4.6.1 Performance Comparison: Reward and Waiting Time

As the three models had been tested with their models modified, a difference in the mean reward was detected with all the three models but not with the waiting time. The maximum mean episode reward (−285,997.42) was obtained with DQN. Q-Learning + SLM was midway on the list and vanilla Q-Learning had the lowest mean of the three. This was the stable ordering in evaluations periods, and this was the major quantitative difference in reward.

Conversely, the three models achieved almost identical results in mean waiting time (around 29.46 seconds) with almost similar standard deviation. One of the key findings was this decoupling of the ranking of rewards and that of waiting-time. This meant that the learning of action distributions that interacted with the shaped reward function dominated the occurrence of reward differences rather than by significant differences in congestion experienced by vehicles. The dominance of DQN Increase Flow may be associated with higher shaped rewards in the states where the flow rate was less than a median value, whereas Q-Learning-based controllers were observed to reward Decrease Flow in states where vehicle-count thresholds prevailed. But the opposing control preference did not result in the ability to produce separable waiting-time results, which led to the question of whether the reward signal was able to differentiate adequately among operationally meaningful policies (Xiao et al., 2024).

The intermediary position of Q-Learning + SLM was in line with its hybrid nature. Occasional gains of SLM directions, enhancing the greedy selection of the Q-table were marginally higher than raw Q-Learning. But the conservative tendency of the SLM to discover Decrease Flow restricted the capability of the model to find the Increase Flow strategy that DQN came to average under operation.

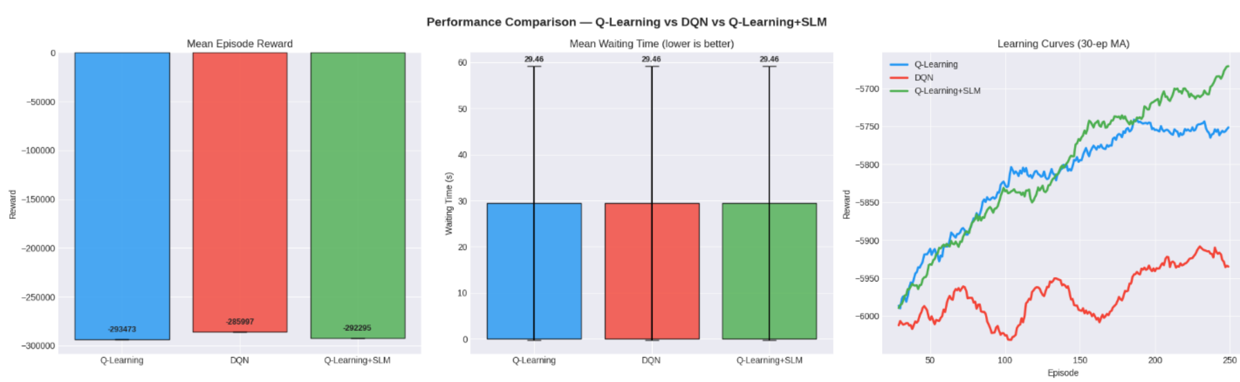

Figure 1: Performance Comparison — Mean Episode Reward, Mean Waiting Time, Learning Curves 30-ep MA

4.6.2 Learning Curve Comparison

Moving-average learning curves of 30 episodes showed distinct dynamics of training. Following step capping, Q-Learning followed a regular increasing pattern as would be expected with tables as to tabular TD convergence with equal episode lengths. DQN exhibited a less steep curve which later began oscillating around the middle of training, which agrees with insights on replay-buffer saturation and policy collapse. The path of Q-Learning + SLM was in between the other two but had greater variance: the standard deviation of the final 100 episodes of the learning curve was 86.74 when compared to 78.63 of Q-Learning.

Going by this increase in variance, it was plausible due to SLM interventions. As soon as the SLM bypassed the preference of the Q-table, it added action decisions not chosen by the table and could not have rated these choices highly. Later Q-updates modified such values of state-action using observed rewards, but recurrent discrepancies between the preference of the SLM and greedy choice of the table provided conflicting update messages and heightened inter-episode variation. This effect correlates with the established principles of trade-offs in hybrid RL in which external advice may assist one part of the state space but causes noise in other parts (Hu et al., 2025). It is interesting to note that all of the three models were approximately 80 percent of their ultimate mean reward by approximately episode 19, which implies early learning rate is similar, later stage behavior distinguishes the ultimate stability and performance.

Figure 2: Training Curves — All Three Models: Episode Rewards, TD Error, Epsilon Decay

4.6.3 Action Preference Comparison as a Policy Diagnostic

Among the learned policies, action distributions offered the most convenient contrast in behavior. Q-Learning was highly focused on Decrease Flow (89.8%) and Maintain at 4.8% and Increase Flow at 5.4%. Q-Learning + SLM had the same directional tendency as the advisory distribution followed by SLM in training: 69.9% Decrease Flow, 24.1% Maintain, and 6.0% Increase Flow. The alignment proposed that repeated SLM guidance was imbibed into the value estimates of the Q-table and was not an external overlay.

DQN broke drastically by breaking down to Everything to Increase Flow. This difference could not be explained only by exploration schedule; more logical explanation was to be found in the interaction between the functions approximation and reward shaping. The continuous representation of DQN could have conditioned to maximize Advance Flow of sampled states since the sampled states were below the flow-rate median ambiguity in the reward function. By contrast, tabular techniques reformed the state space with such a large number of continuous states that they could be reduced to bins with Decrease Flow proving to be the optimal choice owing to threshold vehicle counts. This policy difference between the use of tabular and neural Q-learning based on discretization and shaping interactions is not a new phenomenon and can lead to the opposite action resulting in the same aggregate waiting-time effect (Swapno, 2024).

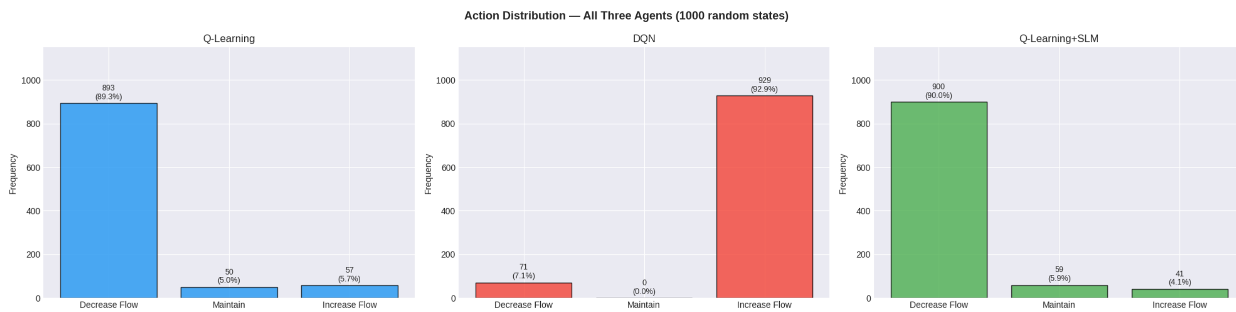

Figure 3: Action Distribution — All Three Agents, 1000 random states

4.7 Discussion: Why Evaluation Didn't Improve

Despite the training changes and the implementation of SLM-based decision-making, none of the agents had a statistically significant decrease in mean evaluation waiting time compared to the baseline; they all stabilized at around 29.46 seconds. This was explained by using five explanations.

To start with, Q-Learning encountered training evaluation horizon discrepancies. It also trained on 200 step episode for grading but was graded on complete rollouts; training curve improvements may therefore represent stability in the capped horizon, as opposed to long-horizon control. Comparison would involve aligning horizons or normalization by the length of the episodes to arrive at conclusive evidence (Louw et al., 2022).

Second, the reward signal might not have been discriminative in a manner that is sufficiently across actions. Under a dominant application of an aggregate measure (total waiting time), when the difference in responses of various actions is insignificant, the spread of Q-values will come to similar estimates and minor initial benefits can bootstrap to the degeneration of single-action dominance. This has been reiterated in traffic RL where the ability to control choices is not explicitly differentiated with the aid of reward shaping (Bouktif et al., 2023).

Third, the environment could have been weakly disposed in response to the action regime put under the experimented demand regime. When the Decrease, Maintain, and Increase Flow generate weakly correlated downstream queue trajectories when the exogenous traffic patterns, then no algorithm can find substantially different policies due to weak discrimination of policy feedback. A direct diagnosis would be to test three constant base policies (laws Directly Decrease and Directly Maintain and Directly Increase) with the same evaluation horizons to identify the variance in the waiting time.

Fourth, the SLM of Q-Learning + SLM was rule based and thresholds to be selected by the rule engine were determined using the same data employed by RL agents. Consequently, SLM guidance was associated with the reward signal the Q-table was already maximizing. The advisor thus was found to have confirmed existing preferences more frequently than it challenged thus limiting its effect on a wait time. A better SLM that is trained on external traffic knowledge may offer new guidance that is not learned in dataset derived shaping.

Fifth, the replay buffer created by DQN rapidly became repeated with homogeneous behavior. After Increase Flow transitions dominated the buffer, other action gradients were expired and therefore it was hard to recover without diversification, as enabled by increased flow incompatibility like prioritized replay or explicit constraints of behavioral diversity (Hu & Li, 2024).

4.8 Limitations and Threats to Validity

Causal interpretation was limited by several factors. The magnitude of rewards differed between training and evaluation because of horizon mismatch and could not be compared without normalization. It also made possible by the increase in action dominance during the post-change (Decrease Flow in both Q-Learning models, Increase Flow in DQN) that simply increasing flow might be contributing to apparent stability by implying homogeneity of behavior rather than assigning a policy that was genuinely adaptive, which is a typical trap in evaluation of traffic RL (Xiao et al., 2024). Initialization was also random and especially among results using DQN any initial trajectories in the replay-buffer can drive the network to other local optima. Specifically: using a fixed seed (42) may cause the network to drift to other local optima. Moreover, there were not many check-ups on the model (not more than 10 evaluation episodes); more check-ups (e.g., 50-100) would make it less unpredictable.

In the case of Q-Learning + SLM, three additional limitations were used. (i) Static percentile scales could not make adaptations to changes of distribution. (ii) The overriding rate was conditioned on the confidence level: by reducing the index, the influence of the SLM would be greater (which may aggravate the stability effect), whereas raising it would render the effect of the SLM as trivial as possible. (iii) SLM is biased toward being conservative (69.9 percent Decrease Flow) could also have prevented finding Increase Flow strategies which are rewarded more by DQN.

4.9 SLM Integration Analysis

4.9.1 Consultation Frequency and Override Behavior

The SLM was checked 15,100 times across 250 training episodes with a maximum number of steps being 200, which confirms that the ten-step trigger was used as intended. This amounted to an average of about 60.4 consultations per episode, and 200 step full episodes with 20 points in them, also had 20 consultation points (steps 10-200).

About 2,507 events or 16.6% of the consultations resulted in the SLM making a hard override (confidence $\geq$ 0.80). In the rest 83.4 it was a mix of probabilities in the system, instead of mandating the recommendation. This behavioral pattern was found to be consistent: confidence of above 0.80 was almost exclusively partly when three or more congestion signals were on at once: this only in a subset, not in most states. The commonalities between the override rate further explained the reason one-on-one Q-Learning + SLM was not as stable as vanilla Q-Learning. That the advisor made roughly a sixth of the decisions presented an element of introducing actions that might have been less appreciated by the Q-table. Every override caused a Q-update, which drew values to the advice of the SLM; when the advisor and table agreed, the cost of stabilizing was very low, whereas frequent re-occurrence of such situations created conflicting update forces and amplified reward variance.

4.9.2 SLM Action Distribution and Absorption into the Q-Table

Among 15,100 consultations, the SLM suggested Decrease Flow in 69.9%, Maintain in 24.1% and Increase Flow in 6.0%. The low Increase Flow share design Increase Flow had been suggested to only increase during obviously under-utilized cases (which were less common in the data).

Directional compensation of this advisory distribution to the final Q-Learning + SLM evaluation action distribution demonstrated that Q-Learning + SLM was dominated by Decrease Flow as vanilla Q-Learning had been dominated by Decrease Flow of 89.8%. This implied it was assumed that repetitive advisory information was involved in Q-table estimates, and not as the occasional outside control.

A key implication followed. Making a new strategic policy direction was not the main contribution of the SLM in this experiment and giving a justification layer that is interpretable to the decisions was instead a contribution made by the SLM to those decisions that the reward signal already stimulated. This added value may be significant in the situations when accountability and the trust of the operators are crucial without the performance improvement. In operation strictly based on performance, this complexity addition and a modest decrease in stability can be more difficult to defend.

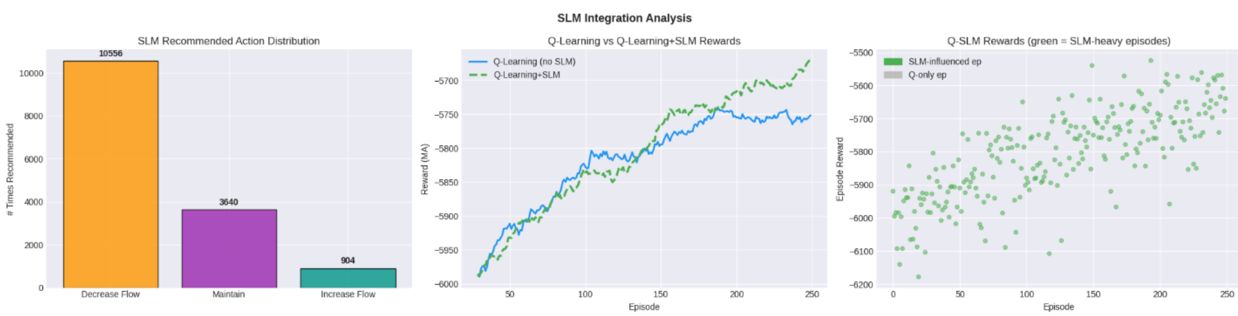

Figure 4: SLM Integration: Recommended Action Distribution, Q-Learning vs Q-Learning + SLM Rewards, Q-SLM Episode Scatter

4.9.3 Interpretability Output: The SLM API in Practice

Every consultation generated four products, namely label, explanation, recommendation, and confidence. To illustrate, when vehicle count was 110, speed was 8.5 km/h, lane occupancy was 92%, flow was 620 and waiting time was 85 seconds in a state with vehicle count of morning and hour = 1, the congestion signals were all on: count was greater than the 67^{th} percentile threshold, speed was less than the 33^{rd} percentile threshold, the occupancy was greater than the 67^{th} percentile threshold, and flow was less than the median. This state was described using the label HEAVY CONGESTION, with the aid of the Decrease Flow with the confidence of 0.90, and a natural-language explanation of how these indicators were connected to the congestion measure and selected intervention.

This audit trail lacked in vanilla Q-Learning and DQN. Q-tables and Q-value vectors are numerically inspectable, but do not give human-readable explanations based on traffic indicators. When the reasoning was logged at every tenth step, the SLM was able to give readable decision traces at approximately a fifth the control points when performing evaluation rollouts, and this inability to sacrifice its support of accountability in the areas of automated traffic control that demand justifications (Hu et al., 2025).

4.9.4 Model Complexity and the Interpretability–Performance Trade-off

The three models were placed at varying levels of a complexity-interpretability spectrum. Q-Learning was simplest and the Q-table was used with 1,892 state entries and three action values of each state. DQN was the most computationally complicated and had 4,739 network parameters and three fully connected networks including replay buffer that can store up to 5,000 experience tuples. Computationally, poorly defined behaviorally, but best explained with the least interpretable, Q-Learning + SLM was an intermediate between them, with a Q-table of 1,863 states (with periodic SLM consultations).

The interpretability was associated with costly stability. The final-window standard deviation of Q-Learning + SLM (86.74) was nearly ten percent bigger than that of Q-Learning (78.63), and the final-window mean reward was lower than DQN and higher than vanilla Q-Learning. The architecture was therefore some sort of a trade-off: it was a little less stable than tabular Q-Learning and received lower rewards than DQN, but at least, offered visibility of the decision. The results showed that the language-informed advice and explanation can be incorporated into tabular RL without significant overhead, implying a viable middle ground between conventional safety-critical infrastructure entailing and language-interpretable settings where interpretability is an initial requirement.

CHAPTER 5: CONCLUSION

5.1 Chapter Overview and Research Recap

The chapter wraps up the study by summarizing the main findings of the study, theoretical and practical contributions, and research areas to be used in the future. The hypothesis under discussion pertained to the possibility and efficiency of reinforcement learning (RL) methods applied to controlling traffic lights in the intersection of an urban intersection. It also examined how RL-agents could be used to achieve better congestion and vehicle waiting times compared to either conventional or similar lightly actuated signal plans. The experiment utilized a simulation-based comparison of three types of controllers namely tabular Q-Learning, Deep Q-Networks (DQN), and a hybrid Q-Learning agent with a Small Language Model (SLM) advisor. The study also sought to control the behavior of policies by isolating behavior and learn reward characteristics, the existing relationship between interpretability of controllers and performance stability.

5.2 Methodological Summary

The study had an evaluation-first approach where signal control was concluded as a problem of episodic RL. A traffic simulation environment that encodes the intersection on a structured state space using traffic indicators including queue length, mean speeds, time of the current signal phase and time of remaining signal phase, was used to model the intersection. Three major control options in the action space were discrete and included Decrease Flow, Maintenance Flow, and Increase Flow that show the phase-time corrections.

The agents were exposed to a gym-like environment connected with a traffic simulator where they were taught policies in a sequence of episodes. Measurements of performance were based on standard RL performance measurements (i.e. cumulative reward) as well as operational measurements (i.e. mean vehicle waiting time). Stability evaluation was specifically focused on training and reward variance evaluation alongside action distribution. The hybrid controller had a periodic advisory element driven by language model which identified the level of congestion, prescribed an action and optionally, provided agent override option with a given confidence threshold. Through this arrangement, interpretability versus stability of control could be measured in RL-based traffic signal operations.

5.3 Key Findings and Interpretation

Among the most notable findings was that though there was a high degree of variation in policy structures, action frequencies and learning dynamics; all three controllers came close to the same mean control vehicle waiting times. This implies that given a very limited and single-intersection setting, different policies can give the same functionally equivalent performance where the reward design is not very fine-grained or sensitive to what would otherwise be different fine-grained state variations.

Q-Learning also demonstrated significant stability in the training process when the episode was step-capped i.e. it had a fixed number of simulation steps per episode. This intervention reduced the variance of rewards drastically, which shows that there had been fewer disruptive outliers and policy convergence increased. Nevertheless, this step cap also had an impact on the learned behavior: the policy was excessively speculative according to the maximization of short-term rewards. Thus, the agent heavily chose the Decrease Flow action even in those states where the phase extension or maintenance could be more realistic. This shortsighted conduct exhibited an important trade-off that step-capping enhanced the quality of control at short periods but at the expense of the quality of long-term control because of the horizon-mismatch between training and evaluation.

DQN on the other hand had the problem of constant policy breakdown. However, despite the methods of exploring decay and learning based on experience, the end policy reversed to choosing a single action, which is the Increase Flow, in most states. This brought the issue of the expressiveness of the state representation and the reward signal, which might have not distinguished the utility of alternative actions. It was also the resultant symptoms of the shortcomings of deep learning in a small-size setting with discrete actions counts; a complex architecture is invariably not connected to superior policies when the input signals themselves are under-informative or the reward detail is unsaturated.

The hybrid controller Q-Learning with an SLM-based advisor had meaningful interpretability improvements. The hybrid model also is able to generate readable decision traces, e.g. a state was identified to be HEAVY CONGESTION, and an action was proposed with a quantified confidence level, unlike the standard agents that merely return Q-values without considerations of the semantic context. These recommendations were made readable by human beings and were used to make the decision-making process of an agent transparent and may be used to help engineers to audit control logic. Such transparency had side effects, however. Introduction of outside advice added variability to the training procedure and the slotted thresholds of the advisor might inculcate jumble or erratic conduct unless exquisitely fined. Moreover, the hybrid controller failed to advance the waiting time of vehicles, although the process became more interpretable, which confirms once again that interpretable control may not be effectively superior to operational control.

The further implication of these results is that the ranking of rewards could be different as compared to a performance ranking. The better the policies are rated on accumulating rewards the worse the outcomes they may have in reality as long as the reward structure is not one that has operational goals as a direct expression. Thus, reward alone is not the appropriate method to issue controller evaluation. To have a clearer view of the policy quality, support metrics such as action diversity, phase transitions, or raw waiting time are required.

5.4 Contributions and Practical Implications

This thesis contributes comprehensively in several ways. To begin with, it shows that small, comprehensible, and evaluation-focused diagnostics such as action distributions or waiting-time profiles are critical companions to reward curbs in traffic signal management. This way, it offers a stricter criterion of assessing RL-sourced signal policies even when there are other rewards than those present in abstract rewards.

Second, the thesis demonstrates that any set of seemingly optimal policies can cause no differences in performance on small-scale. Under these conditions, stability, safety, or interpretability derived metrics can be the core to the difference between two controllers. This observation facilitates the notion that RL policy formulation of infrastructure regulation must not be considered in a vacuum but instead within the scope of deployment limits, societal responsibility, and practical variation.

Third, the hybrid controller has been proven to exhibit a new design of inserting understandable decision logic into previously black-box policies. The advisory system demonstrates that it is possible to retain a high level, explainable, signal layer, which communicates with a learning agent without completely controlling its free will. This can be especially relevant to areas where the factors of auditability and explainability hold as much significance as the raw performance, including the public infrastructure, health, and aviation.

5.5 Limitations and Threats to Validity

Some of the most important factors that limit the generalizability of the study are as follows. First, the test was made on a simplified model of one intersection. Although this could be useful in isolating the analysis of policy behaviors, in the real world of urban traffic systems, intersections are interdependent, and display feedback effects such as spillback or coordinated platoons.

Second, minimal representation of the states was done to minimize overfitting and maximize generality. Nonetheless, this compression might have eliminated important information that may be required to differentiate between the subtle control situations resulting in convergence between policies even in cases where the options of action differed.

Third, the reward feature adds the elements of raw delay and shaping, such as immediate penalty thresholds. Although this method helps to promote quicker learning, it can have concealed any meaningful distinctions among policies and cause the dissociation of reward and operation outcomes. To be implemented in the future, work should be based on multi-objective reward designs which can better represent real trade-offs in the work process, e.g. the trade-off between throughput, safety and fairness.

Fourth, the process of training that employed the step-capping method artificially decreased learning horizons, but evaluation episodes stayed longer. This discrepancy must have provided an induction of behavior which was optimal when training was done but not optimal in deployment or training conditions. These horizon mismatches may lead to distortion of policy quality as well as generalization.

Lastly, the SLM advisor had a pre-determined threshold and regular stimuli. These decision points which are hand-crafted might not be generalized across control situations and patterns of traffic. The injected advice might cause variance or bias when there is weak calibration of the model or the control logic does not incorporate/implement the recommendations of the advisor in a holistic manner.

5.6 Recommendations and Future Work

Future studies should aim at improving matching the training state and operation in deployment. This involves ensuring that the temporal horizon of the training and testing is equated or that the performance measures are adjusted such that they are meaningful between various lengths of the episodes.

Another area that is highly needed to improve is reward design. Multi-objective algorithm of reward that encompasses both queue length, throughput, delay variance, and fairness should be used in the future by controllers. Reward shaping and empirical validation should be increasingly subject to criticism and validation on the basis of operational measures.

It is reasonable to take the experiments to multi-intersection networks as a logical step. Complex coordination issues are posed by such networks and demand analysis of the strength and scalability of RL-based controllers. Graph reinforcement learning techniques, hierarchical control, or distributed multi-agent systems techniques may be useful in overcoming these difficulties.

At interpretability, the SLM-based advisor may be further enhanced through the training of its thresholds with supervised data or dynamic adaptation of the decision triggers. Such interpretability mechanisms could be integrated into deep models, such as through attention mechanisms, counterfactual explanations, and so forth, which would better the results and their reliability.

Lastly, there is open question on robustness to real world variation. Sim-to-real stress testing, which presents controllers with incident conditions, sensor malfunctions, and demand oscillations, should be included in work in the future. This will prove essential to real world implementation where variability and uncertainty form a part and parcel.

5.7 Closing Statement

This thesis concludes that reinforcement learning has a potential to facilitate adaptive traffic signal control though there are major challenges. On simplified models with several control policies, it can seem equally effective in minimizing congestion, again demonstrating the importance of improved reward structure, more realistic state modeling and interpretable evaluation policies. Advised or explanatory layers on a controller can be promising in enhancing transparency and trust, particularly where safety issues are involved. Nevertheless, stability and performance should be weighed responsibly against them. To improve the application of RL-based traffic control to a live urban network, dependence on scalability, robustness, and operational realism must be considered in the future work.

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